

Chemistry Exam 15: Chapters 1-9 (Full Syllabus), Detailed Marking Report

Seemab Ali | Class 9 | Federal Board (FBISE) | Paper Version 4 | Attempt date: 2026-07-14 | Marked: 2026-07-14

PERFECT SCORE: 65/65 (100%)

First-ever perfect score in Chemistry. Fourth perfect score overall, joining Math exam-01, Biology exam-02 re-sit, and Physics exam-03.

12/12

SECTION A (MCQS)

33/33

SECTION B (SHORT)

20/20

SECTION C (LONG)

65/65

TOTAL, 100%

A+

GRADE

SECTION A: Multiple Choice Questions (12/12)

#	Question summary	Correct	Seemab	Result	Marks
1	Branch of chemistry for large-scale manufacture	B: Industrial chemistry	B	Correct	1
2	Homogeneous mixture identification	C: Salt dissolved in water	C	Correct	1
3	Relative atomic mass from 2 isotopes (20 amu 90%, 22 amu 10%)	A: 20.2 amu	A	Correct	1
4	Heisenberg uncertainty principle: cannot determine	B: position and future path	B	Correct	1
5	Lanthanides and actinides belong to	D: f-block	D	Correct	1
6	Dipole-dipole forces act between	B: polar molecules	B	Correct	1
7	Moles in 9.2 g of sodium (Na=23)	B: 0.4 mol	B	Correct	1
8	Molecules in 0.25 mol H ₂ O	A: 1.51×10^{23}	A	Correct	1
9	Oxidation number of Mn in MnSO ₄	A: +2	A	Correct	1
10	Enthalpy change: bonds broken 690, bonds formed 810	B: -120 kJ	B	Correct	1
11	At equilibrium, concentrations of reactants/products	C: remain constant	C	Correct	1
12	Down Group 1, reactivity of elements	B: increases	B	Correct	1

Section A: 12/12, a completely clean sweep of the bubble panel. This extends Chemistry's perfect-MCQ record and closes out the single Q12 miss from exam-14 (equilibrium requires a closed container: correctly identified this time).

SECTION B: Short Questions (33/33)

Q.2(i), MAIN: Moles in 22.2 g of CaCl₂**3/3**

Molar mass CaCl₂ = 40 + 2(35.5) = 111 g/mol. Moles = 22.2 / 111 = 0.2 mol. Method and final number both fully correct.

Only a labelling slip: the final line reads "0.2 g/mol" instead of "0.2 mol". The number and the working are exactly right, so no marks are lost, but this is worth a quick habit fix (see Areas to Watch below).

Q.2(ii), MAIN: Molecules in 0.4 mol of CO₂**3/3**

Molecules = moles x N_A = 0.4 x 6.022x10²³ = 2.4088x10²³ molecules. Correct formula and correct arithmetic (matches the key's 2.409x10²³ to rounding). A first line was crossed out and cleanly redone, the final answer is right.

Q.2(iii), MAIN: Molar mass of Ca(NO₃)₂**3/3**

40 + 2(14 + 3x16) = 40 + 2(62) = 40 + 124 = 164 g/mol. Fully correct, matches key exactly.

Q.2(iv), MAIN: Percentage by mass of nitrogen in NH₄NO₃**3/3**

Molar mass = 14 + 4(1) + 14 + 3(16) = 80 g/mol. Mass of N = 2x14 = 28 g. %N = (28/80) x 100 = 35%. Every step shown and correct.

Q.2(v), MAIN: Oxidation numbers of underlined atoms**3/3**

(a) S in SO₃: x + 3(-2) = 0, x = +6. (b) Cr in CrO₃: x + 3(-2) = 0, x = +6. (c) P in P₂O₅: 2x + 5(-2) = 0, 2x = +10, x = +5. All three correct, including a self-caught arithmetic slip on part (c) that was crossed out and correctly redone to +5.

Q.2(vi), MAIN: Mole ratios for 2Al + 3Cl₂ to 2AlCl₃**3/3**

Mole ratio Al:Cl₂:AlCl₃ = 2:3:2. (a) Cl₂ = (3/2) x 0.4 = 0.6 mol. (b) AlCl₃ = (2/2) x 0.4 = 0.4 mol. Both parts correct.

Q.2(vii), OR: Relative atomic mass of element R (isotopes 10 amu 20%, 11 amu 80%)**3/3**

Ar = (10x20 + 11x80)/100 = (200 + 880)/100 = 1080/100 = 10.8 amu. Correct, matches the answer key exactly.

Q.2(viii), MAIN: State the Heisenberg uncertainty principle**3/3**

"It states that it is impossible to determine, at the same time, both the position (the location) and future path (momentum) for the electron. Therefore a definite orbit for the electron cannot be drawn, we can only map the region where the electron is likely to be found (in an orbital)." A precise, textbook-level statement covering both required points.

Q.2(ix), OR: Carbon dating**3/3**

"Carbon dating estimates the age of once-living material by comparing the C-14 activity of the sample with that of living tissue. When C-14 decays to N-14 the proton number increases from 6 to 7 (a neutron becomes a proton) and a beta particle (electron) is emitted." All three key points present: the dating principle, the proton-number change, and the particle named.

Q.2(x), OR: Down Group 1, reactivity, density, melting point**3/3**

(a) Reactivity increases because the outer electron is farther from the nucleus and is lost more easily. (b) Density increases. (c) Melting point decreases. All three predictions correct, with the reason given for reactivity.

Q.2(xi), OR: Why solid NaCl does not conduct but molten/aqueous NaCl does**3/3**

"In solid NaCl the ions are locked in fixed positions in the lattice and cannot move, so it does not conduct. When molten or dissolved, NaCl ionizes and the ions move freely, so it conducts. Aqueous acids conduct because they ionize in water to give freely moving [ions]." Complete answer, including the acids sub-question.

Section B: 33/33. Every one of the 11 required questions was attempted and every sub-part completed, including questions that have historically been the source of Seemab's most common deduction (leaving a sub-part unfinished). None of that here.

SECTION C: Long Questions (20/20)**Q.3(i), MAIN: Relative atomic mass of uranium; alpha decay of U-238****5/5**

(a) Relative atomic mass defined: "the average mass of the isotopes of an element compared with 1/12 the mass of a carbon-12 atom." $Ar(U) = (99.27 \times 238 + 0.72 \times 235 + 0.006 \times 234) / 100 = (23626.26 + 169.2 + 1.404) / 100 = 23796.864 / 100 = 237.97$ amu.

(b) $238\text{-}92\text{-U} \rightarrow 234\text{-}90\text{-Th} + 4\text{-}2\text{-He}$. The particle emitted is an alpha particle (helium nucleus). Both parts fully correct with every arithmetic step shown and reconciled.

Q.3(ii), OR: Empirical and molecular formula (82.7% C, 17.3% H, M=58 amu)**5/5**

(a) C: $82.7/12 = 6.89$. H: $17.3/1 = 17.3$. Divide by 6.89: C = 1, H = 2.51, then x2 to clear the fraction: empirical formula = C_2H_5 .

(b) Empirical mass = $2(12) + 5 = 29$. $n = 58/29 = 2$. Molecular formula = $(C_2H_5)_2 = C_4H_{10}$ (butane). Both steps correct and match the answer key exactly, including the naming of the product as butane.

Q.3(iii), MAIN: Enthalpy change for $N_2 + 3H_2$ to $2NH_3$ from bond energies**5/5**

Formula: $\Delta H = (\text{sum of bonds broken}) - (\text{sum of bonds formed})$.

Bonds broken (reactants): $1(N \text{ triple-bond } N) + 3(H-H) = 945 + 3(436) = 945 + 1308 = 2253$ kJ.

Bonds formed (products): $2NH_3 = 6(N-H) = 6(391) = 2346$ kJ.

$\Delta H = 2253 - 2346 = -93$ kJ/mol. Negative, so the reaction is exothermic.

Every line matches the answer key precisely, including the correct exothermic conclusion.

Q.3(iv), MAIN: Propane combustion stoichiometry (4.4 g propane)**5/5**

Molar mass $C_3H_8 = 3(12) + 8(1) = 44$ g/mol.

(a) Moles $C_3H_8 = 4.4/44 = 0.1$ mol.

(b) $O_2:C_3H_8 = 5:1$, $O_2 = 5 \times 0.1 = 0.5$ mol.

(c) $CO_2:C_3H_8 = 3:1$, $CO_2 = 3 \times 0.1 = 0.3$ mol; mass = $0.3 \times 44 = 13.2$ g.

(d) $H_2O:C_3H_8 = 4:1$, $H_2O = 0.4$ mol; molecules = $0.4 \times 6.022 \times 10^{23} = 2.409 \times 10^{23}$.

All four sub-parts (a) through (d) correct, a clean run through a four-step mole-ratio chain.

Section C: 20/20. All four long questions, both the numerical mole-chain problems and the bond-energy/nuclear-chemistry theory questions, were completed with full working and correct final answers.

Deduction Log (reconciles to headline total)

Section	Marks available	Marks lost	Reason	Marks awarded
A: MCQs	12	0	None	12
B: Short questions (11)	33	0	None	33

C: Long questions (4)	20	0	None	20
Total	65	0	Zero deductions across the entire paper	65

Strengths

- Perfect MCQ panel, 12/12, including the exact equilibrium concept (closed system) that cost the single mark on exam-14
- Isotope-weighted relative atomic mass calculations correct on both the 2-isotope MCQ and the 3-isotope Section C uranium question, with every multiplication and addition line shown
- Nuclear equation for alpha decay balanced correctly on both mass and charge, particle correctly named
- All three oxidation-number sub-parts (S, Cr, P) solved correctly, including catching and correcting her own arithmetic slip mid-working
- Empirical to molecular formula chain (percentage composition to mole ratio to simplest formula to molecular formula) executed flawlessly, ending with the correct compound name (butane)
- Bond-energy enthalpy calculation fully correct with the right sign convention and the correct exothermic/endothermic conclusion
- Four-part propane combustion stoichiometry chain (moles to O₂ to CO₂ mass to H₂O molecules) completed without a single slip
- Every Section B and Section C question attempted in full, no sub-parts skipped, directly answering the "stopping before finishing every sub-part" pattern flagged on several recent papers

Nothing to Fix, One Habit to Watch

- No marks were lost anywhere on this paper.
- Q.2(i): the final line was labelled "0.2 g/mol" when the quantity is actually in mol (moles, not molar mass). The number itself is correct and no mark was deducted, but it is worth a quick habit check: after calculating moles, write the unit as "mol", and only use "g/mol" for a molar mass.

Comparison vs Previous Chemistry Attempts

Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
12 Apr 2026	exam-01-ch1-2	12/12	31/33	21/20*	64/65	98.5%	A+
17 Apr 2026	exam-02-ch3-5	12/12	31/33	16.5/20	59.5/65	91.5%	A+
12 May 2026	exam-03-ch6	12/12	30.5/33	15/20	57.5/65	88.5%	A+
12 Jun 2026	exam-04-ch7	12/12	30/33	18/20	60/65	92.3%	A+
24 Jun 2026	exam-12-ch1-9	12/12	31.5/33	19/20	62.5/65	96.2%	A+
30 Jun 2026	exam-13-ch1-9	12/12	31.5/33	17/20	60.5/65	93.1%	A+
6 Jul 2026	exam-14-ch1-9	11/12	31.5/33	17.5/20	60.0/65	92.3%	A+
14 Jul 2026	exam-15-ch1-9 (this exam)	12/12	33/33	20/20	65/65	100%	A+

* Chemistry exam-01 Section C shows 21/20 due to a historical bonus mark in the original marking (see cumulative report footnote).

This is Seemab's best Chemistry result to date and her first perfect score in the subject, beating the previous high of exam-01 (98.5%). It closes out a run of three consecutive full-syllabus Ch1-9 papers (exam-12: 96.2%, exam-13: 93.1%, exam-14: 92.3%) with the strongest of the four, and it is also the first Chemistry paper on which Section B and

Section C were both perfect at the same time.